

A further conjecture on OEIS A083178, settled by the proof already in hand

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Abstract

OEIS A083178 carries more than one conjecture. One of them has a proof; this paper settles a DIFFERENT one, and it needs no new model and no new engine – only the linear recurrence already established for the sequence. A claim about a C-finite sequence whose own characteristic polynomial is a multiple of the established one is decided by polynomial divisibility, and where the claim is an identity rather than a recurrence, by divisibility together with a finite window of agreement.

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1 The premise

The entry's first terms are 1, 2, 3, 22, 32, 33, 322, 332, $\dots$, with offset 1. The sequence satisfies

$$a(n) = 11 a(n-3) - 10 a(n-6),$$

a linear recurrence of order 6 with characteristic polynomial

$$q(x) = x^6 - 11x^3 + 10.$$

This recurrence is proved in this project; it is the premise of everything below and nothing else is assumed. It was confirmed against every term the entry publishes from the index where it begins to hold, with at least $6 + 2$ further terms after that index.

2 The further conjecture

Separately from the conjecture already settled, the entry records

Conjecture: $a(n) = 10*a(n-3)+a(n-6)-10*a(n-9)$ for $n>10$. - _Colin Barker_, Oct 14 2014

The further claim is another RECURRENCE, so the claim itself is that a satisfies the recurrence whose characteristic polynomial is p . Since $q \mid p$ and a satisfies the recurrence of q , it satisfies that of p , which is exactly what is claimed. No comparison of terms is needed, and none is made.

Theorem 1. *The conjecture displayed above is true.*

3 What is and is not claimed here

This is a second result on an entry that already has one, and it is a second result only because it is a DIFFERENT statement. A generating function whose denominator's reciprocal is exactly q says precisely what the proved recurrence says, in another notation; such restatements are identified and are not counted. The conjecture settled here is not of that kind.